

Research Interests

- Post-Quantum Cryptography (PQC) and Lattice-based Cryptography
- Fully Homomorphic Encryption (FHE), including Threshold FHE
- Privacy Enhancing Technologies (PETs)

Education

- **Ph.D. in Mathematical Sciences—Cryptography**, *Dept. of Mathematical Sciences, Seoul National University (SNU), Korea*, Sep. 2019–Feb. 2025
 - **Adviser:** Prof. Jung Hee Cheon
 - **Thesis Title:** Accelerating Homomorphic Computation through Machine-Efficient Arithmetic
- **B.S. in Mathematical Sciences**, *Dept. of Mathematical Sciences, SNU, Korea*, Mar. 2013–Aug. 2019

Selected Publications

Selected recent publications. The complete list is available in the Publications section.

- Hyeongmin Choe, Jaehyung Kim, Damien Stehlé, Elias Suvanto, “Leveraging Discrete CKKS to Bootstrap in High Precision,” **ACM CCS 2025** (C06)
- Jung Hee Cheon, Hyeongmin Choe, Minsik Kang, Jaehyung Kim, Seonghak Kim, Johannes Mono, Taeyeong Noh, “Grafting: Decoupled Scale Factors and Modulus in RNS-CKKS,” **ACM CCS 2025** (C05)
- Jung Hee Cheon, Hyeongmin Choe[†], Alain Passelègue, Damien Stehlé, Elias Suvanto, “Attacks Against the IND-CPA^D Security of Exact FHE Schemes,” **ACM CCS 2024** (C03)
- Jung Hee Cheon, Hyeongmin Choe, Julien Devevey, Tim Güneysu, Dongyeon Hong, Markus Krausz, Georg Land, Marc Möller, Damien Stehlé, MinJune Yi, “HAETAE: Shorter Lattice-Based Fiat–Shamir Signatures,” **CHES 2024** (C02)

Awards & Honors

Awards

- **Korean National Cryptography Contest**, *National Security Research Institute (NSR)*. A national annual competition for outstanding cryptography research papers, co-authored by undergraduates to postdocs. Grand/Excellence Prize: best paper from all/each track(s).
 - **Excellence Prize** (for C05), Oct. 2025
 - **Grand Prize** (for C03), **Honorable Mention** (for J07), and **Special Prize** (for J05), Oct. 2024
 - **Honorable Mention** (for M01), Oct. 2022
- **Korean PQC Standardization (Kpqc)**, *NSR and National Intelligence Service (NIS)*. A three-year competition for standardizing Korean PQC Algorithms (Sep. 2021–Jan. 2025).
 - **Winner in KEM/PKE:** SMAUG-T Key Encapsulation Mechanism
 - **Winner in Digital Signature:** HAETAE Digital Signature schemeWebsite: <https://www.kpqc.cryptolab.co.kr>
- **TA Awards**, *Dept. of Mathematical Sciences, SNU*, Aug. 2023
 - **Excellence in Teaching:** for teaching “Honor Calculus Practice 1 (2023 Spring).”
- **2020 iDASH Genomic Data Privacy & Security Protection**, *American National Institutes of Health (NIH)*, Dec. 2020
 - **First Place (Track I):** “Secure Multi-label Tumor Classification using Homomorphic Encryption.” The result was later published as J04.

Honors

- **ACM CCS 2024 Doctoral Symposium Travel Grant**, *ACM SIGSAC*, Oct. 2024
- **BK 21+ Scholarship**, *Ministry of Education of Korea*, Sep. 2019–Feb. 2025
- **Presidential Undergraduate Science Scholarship**, *Korea Student Aid Foundation*, Mar. 2013–Dec. 2018

Experiences

- **Postdoctoral Researcher**, *University of Luxembourg, Luxembourg*, Nov. 2025–Present
 - **Topic:** Broadly on Lattice-based Cryptography, FHE, PQC, and PETs
- **Cryptography Engineer**, *CryptoLab Inc., Korea*, Mar.–Oct. 2025
 - **Topic:** FHE (applications, implementation, and cryptanalysis) and PQC (applications, implementation, and standardization)
- **Research Visit**, *École Normale Supérieure de Lyon, France*, Sep.–Oct. 2023 (during PhD studies)
 - **Topic:** Concrete construction of a new digital signature scheme.
- **Sergeant**, *Intelligence System Management Group, Republic of Korea Air Force (ROKAF)*, July 2015–July 2017 (mandatory military service)

Public & Professional Services

Invited Talks

Conference Invited Talks

- **2025 KMS Spring Meeting**, “HAETAE and SMAUG-T: Korean PQC Standards,” *organized by Korean Mathematical Society (KMS)*, KAIST, Korea, Apr. 25, 2025 (1h)
- **KIAS-JBNU KpqC Workshop**, “HAETAE: Rejecting on Hyperballs,” *organized by Korea Institute for Advanced Study (KIAS)*, Jeonbuk National Univ., Korea, May 19, 2023 (2h)

Seminar Invited Talks

- **Kookmin Univ., Korea**, “Security in the Post-Quantum Era: Post-Quantum Cryptography and Standardizations (Translated),” *at Dept. of Information Security, Cryptography, and Mathematics*, June 13, 2025 (1h)
- **Ruhr Univ. Bochum, Germany**, “Recent Advances in Fully Homomorphic Encryption,” *at Faculty of Computer Science, Security Engineering*, Jan. 21, 2025 (1.5h)
- **Sungshin Women’s Univ., Korea**, “HAETAE: Shorter Lattice-based Fiat-Shamir Signatures,” *at Dept. of Convergence Security Engineering*, May 21, 2024 (1.5h)
- **2024 KpqC Winter Camp**, “HAETAE,” *organized by KpqC Research Group, at Sogang Univ., Korea*, Feb. 27, 2024 (1h)
- **2024 Algebra Camp**, “Bridging Algebraic Number Theory to Post-Quantum Digital Signatures,” *organized by QSMS, at Yangpyeong Bloomvista, Korea*, Feb. 5, 2024 (30m)
- **Korea Univ., Korea**, “HAETAE, a Post-Quantum Signature Scheme,” *at School of Cybersecurity*, July 24, 2023 (2h)
- **2023 KpqC Winter Camp**, “Introduction to HAETAE,” *organized by KpqC Research Group, at Chung-Ang Univ., Korea*, Feb. 22, 2023 (1h)

Invited Lectures

- **Lecture Series at PQC Migration Platform Seminar**, *jointly organized by LG U+, NIA, and CryptoLab Inc.*, Introductory lectures on PQC and lattice-based KEMs and digital signatures, July 25, Sep. 30, Oct. 29, and Nov. 25, 2025 (8h)
- **Sungshin Women’s Univ., Korea**, “HAETAE: Lattice-based Digital Signature (Translated),” *Graduate Course at Dept. of Convergence Security Engineering*, Sep. 23, 2025 (2h)
- **Cryptography Training for Information Security Professionals**, *organized by Korea Cryptography Forum*, Pre-recorded lectures on lattice-based PQC, May–Jun. 2025 (3h)
- **Dongguk Univ., Korea**, “Security in the Post-Quantum Era: Post-Quantum Cryptography and Standardizations (Translated),” *Undergraduate Course, at Dept. of CS & AI*, May 29, 2025 (1.5h)

- **PQC Training Course**, *organized by CryptoLab Inc.*, Two half-day lectures on the concrete security of lattice-based PQC, July 16-17, 2024 (7h)
Material: https://github.com/hmchoe0528/PQC_training.
- **2nd 10-10 Gauss Distinguished Lecture Series**, *organized by IMDARC, SNU*, Pre-study on Damien Stehlé’s Distinguished Lecture on NIST PQC Standards, Sep. 15, 2023 (0.5h)

Editorial & Academic Services

- **Co-Editor**, CKKS.org, Dec. 2025 – Present.
- **Program Committee Member**: ICISC 2025 and ACM CCS 2026.
- **Journal Reviewing**: (Sub/External) Reviewer for
 - Journal of Cryptology (JoC) in 2023, 2026,
 - Design, Codes and Cryptography (DCC) in 2024–2025,
 - IEEE Transactions on Information Forensics and Security publication information (TIFS) in 2026.
- **Conference Reviewing**: Sub/External reviewer for
 - Asiacrypt 2022, 2025,
 - ACM CCS 2022,
 - FHE.org 2022, 2026,
 - PQCrypto 2023–2024,
 - PKC 2024,
 - Eurocrypt 2024, 2026,
 - Crypto 2026.

Publications

Authors are listed alphabetically by last name, per the AMS 2004 authorship statement, unless marked with an asterisk ().*

A dagger (†) denotes the corresponding author when applicable.

Conferences

- C06 Hyeongmin Choe, Jaehyung Kim, Damien Stehlé, Elias Suvanto, “Leveraging Discrete CKKS to Bootstrap in High Precision,” **ACM CCS 2025** (ACM Conference on Computer and Communications Security).
- C05 Jung Hee Cheon, Hyeongmin Choe, Minsik Kang, Jaehyung Kim, Seonghak Kim, Johannes Mono, Taeyeon Noh “Grafting: Decoupled Scale Factors and Modulus in RNS-CKKS,” **ACM CCS 2025** (ACM Conference on Computer and Communications Security).
- C04 Hyeongmin Choe[†], “Toward Practical Threshold FHE: Low Communication, Computation and Interaction,” **ACM CCS 2024 Doctoral Symposium**. 3-Page Extended Abstract.
- C03 Jung Hee Cheon, Hyeongmin Choe[†], Alain Passelègue, Damien Stehlé, and Elias Suvanto, “Attacks Against the IND-CPA^D Security of Exact FHE Schemes,” **ACM CCS 2024** (ACM Conference on Computer and Communications Security).
- C02 Jung Hee Cheon, Hyeongmin Choe, Julien Devevey, Tim Güneysu, Dongyeon Hong, Markus Krausz, Georg Land, Marc Möller, Damien Stehlé, and MinJune Yi, “HAETAE: Shorter Lattice-Based Fiat-Shamir Signatures,” **CHES 2024** (Conference on Cryptographic Hardware and Embedded Systems).
- C01 Jung Hee Cheon, Hyeongmin Choe[†], Dongyeon Hong, and MinJune Yi, “SMAUG: Pushing Lattice-based Key Encapsulation Mechanisms to the Limits,” **SAC 2023** (Selected Areas in Cryptography).

Journals

- J07 Jung Hee Cheon, Hyeongmin Choe[†], Seunghong Kim, Yongdong Yeo, “Multi-Party Homomorphic Encryption with Dynamicity and Ciphertext Reusability,” **IACR CiC** (Communications in Cryptology), vol. 3(1), May 2026.
- J06 Jung Hee Cheon, Hyeongmin Choe, and Jai Hyun Park, “Tree-based Lookup Table on Batched Encrypted Queries using Homomorphic Encryption,” **JKMS** (Journal of the Korean Mathematical Society), vol. 62(5), pp. 1237–1263, Sep. 2025.
- J05 Jung Hee Cheon, Hyeongmin Choe[†], Jungjoo Seo, Hyoeun Seong, “SMAUG(-T), Revisited: Timing-secure, More Compact, Less Failure,” **IEEE ACCESS**, vol. 12, pp. 188386–188397, Dec. 2024.

- *J04 Seungwan Hong, Jai Hyun Park, Wonhee Cho, Hyeongmin Choe and Jung Hee Cheon[†], “Secure tumor classification by shallow neural network using homomorphic encryption,” **BMC Genomics**, vol. 23, no. 284, Apr. 2022.
- J03 Jung Hee Cheon, Hyeongmin Choe, Donghwan Lee and Yongha Son[†], “Faster Linear Transformations in HElib, revisited,” **IEEE Access**, vol. 7, pp. 50595–50604, Apr. 2019.
- *J02 Siyul Lee and Hyeongmin Choe, “On Fourth-order Iterative Methods for Multiple Roots of Nonlinear Equations with High Efficiency,” **JoCAAA (Journal of Computational Analysis and Applications)**, vol. 18, no. 1, pp. 109–120, Jan. 2015.
- *J01 Siyul Lee and Hyeongmin Choe, “Multiplicational Combinations and A General Scheme of Single-step Iterative Methods for Multiple Roots,” **JoCAAA (Journal of Computational Analysis and Applications)**, vol. 15, no. 6, pp. 1138–1149, Oct. 2013.

Technical Articles & Specifications (non-refereed)

- T04 Hyeongmin Choe, Jeongdae Hong, “Korean Post-Quantum Cryptography Algorithm HAETAE: Lattice-based Digital Signature Scheme,”¹ Invited Technical Article, **Review of KIISC (Korea Institute of Information Security and Cryptology)**, vol. 35, no. 3, pp. 15–20, June 2026.
- T03 Hyeongmin Choe, Jeongdae Hong, “Korean Post-Quantum Cryptography Algorithm SMAUG-T: Lattice-based Key Encapsulation Mechanism,”² Invited Technical Article, **Review of KIISC (Korea Institute of Information Security and Cryptology)**, vol. 35, no. 3, pp. 21–27, June 2026.
- T02 Jung Hee Cheon, Hyeongmin Choe, Julien Devevey, Tim Güneysu, Dongyeon Hong, Markus Krausz, Georg Land, Marc Möller, Junbum Shin, Damien Stehlé and MinJune Yi, “HAETAE: Hyperball bimodal module rejection signature scheme,” Algorithm Specification v0.9–v3.0, along with **KpqC Competition** and **NIST PQC Additional Signatures**.
- T01 Jung Hee Cheon, Hyeongmin Choe, Joongeun Choi, Dongyeon Hong, Jeongdae Hong, Chi-Gon Jung, Honggoo Kang, Janghyun Lee, Seonghyuck Lim, Aesun Park, Seunghwan Park³, Jungjoo Seo, Hyoeun Seong, and Junbum Shin, “SMAUG(-T): the Key Exchange Algorithm based on Module-LWE and Module-LWR,” Algorithm Specification v0.9–v4.0, for **KpqC Competition**.

Manuscripts (non-refereed)

Manuscripts that are archived or near completion.

- M01 Jung Hee Cheon, Hyeongmin Choe, Saebyul Jung, Duhyeong Kim, Dah Hoon Lee, and Jai Hyun Park, “Arithmetic PCA for Encrypted Data,” **Cryptology ePrint Archive, Paper 2023/1544**, Oct. 2023.

Teaching Record

- **Calculus TA Seminar**, at *Dept. of Mathematical Sciences, SNU*, 2024 Spring
 - **Role:** TA, guiding new TAs on teaching skills and student management strategies.
- **Calculus Practice Sessions**, at *College of Natural Sciences, SNU*, 2020–2023 (7 semesters)
 - **Role:** TA and Lecturer, delivering 2-hour weekly practice sessions with summarized content and guided students.
 - **Teaching Evaluation (Student Survey):** Avg. 94.6 / 100. Awarded “*Excellence in Teaching*” in 2023 Spring.
- **(i-TAP) Post-Quantum Cryptography**, at *SK Hynix Inc.*, Apr.–May (5 weeks), 2021
 - **Role:** TA and Co-lecturer, for i-TAP (Innovative Technology Advancement Program), delivering 8 of 26 total hours as a co-lecturer. Also contributed to course material development and led Q&A and discussion sessions on lattice-based PQC.
- **Korean Mathematical Olympiad (KMO) Winter/Summer Schools**, *organized by KMS*, 2013–2014
 - **Period:** Jan. & Aug., 2013, and Jan. & Aug., 2014 (each 2–3 weeks)
 - **Role:** Residential TA, managing and supporting elementary to high school students during intensive camp; delivering exercise sessions.

Patents

The list includes (filed or granted) patents that are publicly disclosed.

¹Title translated. Original title is “한국형 양자내성암호 HAETAE: 격자기반 전자서명 스킴.”

²Title translated. Original title is “한국형 양자내성암호 SMAUG-T: 격자기반 키 캡슐화 메커니즘 스킴.”

- Jung Hee Cheon and Hyeongmin Choe, “Apparatus for conversion of homomorphic encrypted message and method thereof,”
– Korea Patent No. KR102782557, Granted, Feb. 2025.
- Jung Hee Cheon and Hyeongmin Choe, “Apparatus for generating blind signature and method thereof,”
– US Patent No. US12309293, Granted, May 2025.
– Korea Patent No. KR102882499, Granted, Nov. 2025.
- Jung Hee Cheon, Hyeongmin Choe, and Dongyeon Hong, “Electronic device for encrypting data by public key and methods thereof,”
– US Patent No. US12452039, Granted, Oct. 2025.
– Application No. KR20240081407A, and US20260025263A1, Pending.
- Jung Hee Cheon, Hyeongmin Choe, and Jai Hyun Park, “Electronic device for searching encrypted data and methods thereof,”
– US Patent No. US12367404, Granted, July 2025.
– Application No. KR20240118024A, Pending.
- Jung Hee Cheon, Hyeongmin Choe, and Jai Hyun Park, “Electronic device for making decision and methods thereof,”
– Applications No. KR20240118025A and US20240289650A1, Pending.
- Jung Hee Cheon, Hyeongmin Choe, Minsik Kang, and Jaehyung Kim, “Electronic apparatus for performing operations on homomorphic ciphertext and control method thereof,”
– Applications No. KR20250018059A, US20250266984A1, and EP4607842A1, Pending.
- Jaehyung Kim, Hyeongmin Choe, Elias Suvanto, and Damien Stehlé, “Electronic apparatus and controlling method thereof,”
– Applications No. KR20260020889A, US20260046110A1, Pending.

Contributed Talks

Conferences and Workshops

- **ACM CCS 2025**, at *Taipei, Taiwan*, Oct. 14, 2025
– Title: Leveraging Discrete CKKS to Bootstrap in High Precision
- **ACM CCS 2025**, at *Taipei, Taiwan*, Oct. 14, 2025 (jointly with Minsik Kang)
– Title: Grafting: Decoupled Scale Factors and Modulus in RNS-CKKS
- **ACM CCS 2024 Doctoral Symposium**, at *Salt Lake City, US*, Oct. 14, 2024
– Title: Toward Practical Threshold FHE: Low Communication, Computation and Interaction
- **Selected Areas in Cryptography (SAC) 2023**, at *Univ. of New Brunswick, Canada*, Aug. 16, 2023
– Title: SMAUG: Pushing Lattice-based Key Encapsulation Mechanisms to the Limits

Research Camps and Colloquiums

- **KpqC Contest 2nd Round Colloquium**, organized by *KpqC Research Group*, at *Hansung Univ., Korea*, Aug. 28, 2024 (0.5h)
– Title: HAETAE v3.0
- **2024 KMS Spring Meeting**, organized by *KMS*, at *Daejeon Convention Center, Korea*, Apr. 19, 2024 (0.5h)
– Title: IND-CPA^D and KR^D Security of Exact (F)HEs
- **2024 Crypto Winter Camp**, organized by *SNU Cryptography Lab*, at *Vivaldi Park, Korea*, Jan. 4, 2024 (1h)
– Title: IND-CPA^D and KR^D Security of FHE and Application to Threshold-FHE
- **2023 Crypto Winter Camp**, organized by *SNU Cryptography Lab*, at *Konjiam Resort, Korea*, Jan. 5, 2023 (1h)
– Title: Introduction to SMAUG KEM and HAETAE Signature Schemes
- **2022 KMS Spring Meeting**, organized by *KMS*, *Virtual*, Apr. 28, 2022 (0.5h)
– Title: Efficient, Round-optimal Blind Signatures from Standard Assumptions
- **2020 KMS Annual Meeting**, organized by *KMS*, *Virtual*, Oct. 24, 2020 (0.5h)
– Title: Conversion between Two RLWE-based FHE Schemes and its Application